Differentiation: Tangents and Normals

Questions 1–27

Question 1. Evaluate $f'(2)$ for each of the following, leaving your answers in exact form:

a) $f(x) = x^3 - 2x^{-1}$

b) $f(x) = \sqrt{x} + 2x^4$

c) $f(x) = \dfrac{x^5 - 1}{x^2}$

d) $f(x) = \sqrt{x}(1 - x)$

Question 2. For each of the following curves, find the equation of the tangent line which passes through the given point. Give your answers in the form $ax + by + c = 0$ where a, b, c are integers and $a > 0$:

a) $y = x^2$ at $(2, 4)$

b) $y = x^3 - x + 5$ at $(1, 5)$

c) $y = \dfrac{1}{x}$ at $\left(-2, -\dfrac{1}{2}\right)$

d) $y = 2\sqrt{x}$ at $x = 1$

Question 3.

a) Show that the tangent to the curve $y = 3 - x^2$ when $x = 0$ may be written in the form $y - k = 0$ where k is a value to be found.

b) Explain the significance of the gradient of this tangent.

Question 4. For each of the following curves, find the equation of the normal line which passes through the given point. Give your answers in the form $ax + by + c = 0$ where a, b, c are integers and $a > 0$:

a) $y = 3x^2$ at $(1, 3)$

b) $y = 5x^4 - x^2$ at $(-1, 4)$

c) $y = \dfrac{1}{x^2}$ at $\left(2, \dfrac{1}{4}\right)$

d) $y = \sqrt{x}$ at $(1, 1)$

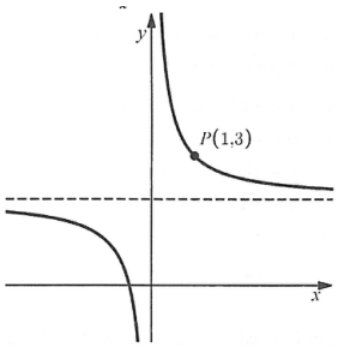
e) $y = x - x^4$ when $x = -1$

Question 5. For the equation $y = 5x^2 + 1$, find the value of x such that $\dfrac{dy}{dx} = 30$.

Question 6. For the equation $y = x^3 - 2x$, find the values of x such that $\dfrac{dy}{dx} = 10$.

Question 7. For the equation $y = x^{1/2}$, find the value of x such that $\dfrac{dy}{dx} = \dfrac{1}{4\sqrt{2}}$.

Question 8. The graph of $y = \dfrac{1}{x} + 2$ is shown below.



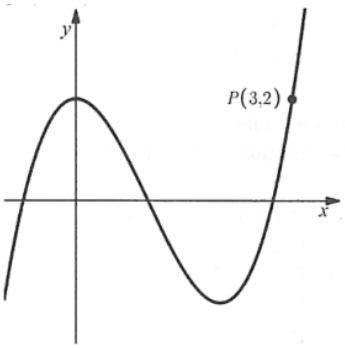
The point P has coordinates $(1, 3)$.

a) Find the equation of the tangent line at P in the form $y = mx + c$ where m and c are integers to be found.

b) Find the equation of the normal line at P in the form $y = mx + c$ where m and c are integers to be found.

c) Find the coordinates where the tangent and normal line at P meets the coordinate axes.

Question 9. The graph of $y = x^3 - 3x^2 + 2$ is shown below.



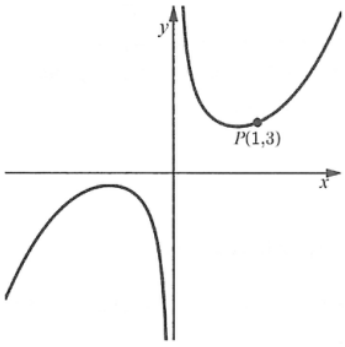
The point P has coordinates $(3, 2)$.

a) Find the equation of the normal line at P in the form $ax + by = c$ where a, b and c are integers to be found and $a > 0$.

b) Find the coordinates where the normal line crosses the x-axis and y-axis.

c) Hence calculate the area of the triangle bounded by the normal line at P and the coordinate axes.

Question 10. The graph of $y = \dfrac{1}{x} + x^3 + 1$ is shown below.



The point P has coordinates $(1, 3)$.

a) Find the equation of the tangent line at P in the form $y = mx + c$ where m and c are integers to be found.

b) Find the coordinates where the tangent line crosses the x-axis and y-axis.

c) Find the coordinates where the tangent meets the curve again.

Question 11. The point A lies on the curve

$$y = 5x^2 - \frac{1}{x}.$$

The tangent line to the curve at point A which lies on the curve is parallel to the line $y + 9x = 5$.

a) Find the coordinates of the point A.

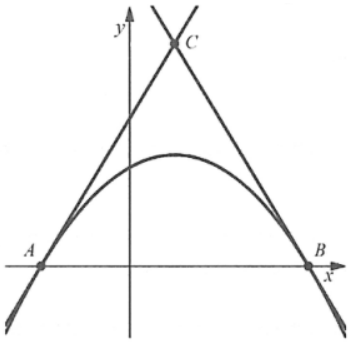
b) Find the equation of the tangent line to the curve at the point A in the form $y = mx + c$ where m and c are integers to be found.

c) Hence find the other point where the tangent intersects the curve.

Question 12. A curve has equation $y = \frac{k}{x}$ where k is a positive constant. Find $\frac{dy}{dx}$ at $\left(\frac{1}{2}, 2k\right)$ giving your answer in terms of k.

Question 13. A curve has equation $y = x^3 + px^2 + qx + 5$. It passes through the point $P(2, 1)$. The gradient of the curve at P is 10. Find the value of p and q.

Question 14. The graph below shows the curve $f(x) = -x^2 + 2x + 8$ which meets the x-axis at points A and B. The tangent lines to the curve at A and B meet at point C. Find the area of the triangle ABC.



Question 15. A curve has equation $y = x^3 + (1+p)x^2 + qx + 5$. It passes through the point $P(-1, 6)$. The gradient of the normal to the curve at P is $\frac{1}{2}$. Find the value of p and q.

Question 16. A curve has equation $y = 3x^3 - 5x^5 + 1$. At $x = -1$ the tangent line has the same gradient as another tangent line to the curve. Find the equation of this other tangent line.

Question 17. Show that the equation of the tangent to the curve $y = \frac{1}{x^2}$ at the point $x = a$ is given by $a^3y+2x = 3a$.

Question 18.

a) Show that the equation of the tangent to the curve $y = x(x^2+2)$ at the point $x = a$ is given by $y = (3a^2+2)x-2a^3$.

b) Hence explain why the gradient of the curve $y = x(x^2 + 2)$ is always positive.

Question 19. Prerequisite knowledge for Q19: Second derivatives A curve has equation $y = x^2(1 - kx)$ where k is a positive constant. Find $\dfrac{d^2y}{dx^2}$ when $x = 2$ giving your answer in terms of k.

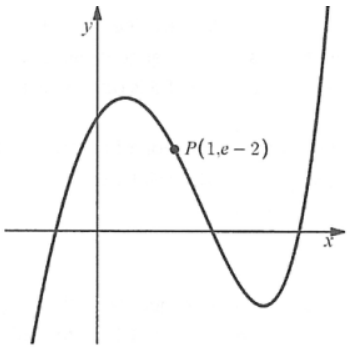
Question 20. Prerequisite knowledge for Q20–Q27: Differentiation of the form ae^{kx} Evaluate $f'(1)$ for each of the following, leaving your answers in exact form:

a) $f(x) = e^{6x}$
b) $f(x) = \sqrt{x} - e^{2x}$
c) $f(x) = x^{4/3} + e^{2x}$

Question 21. Find the equation of the tangent line to the curve $y = e^x$ at $(0, 1)$ in the form $y = mx + c$ where m and c are integers to be found.

Question 22. Find the equation of the normal line to the curve $y = x^2 - e^{2x}$ at $(0, -1)$ in the form $y = mx + c$ where m and c are integers to be found.

Question 23. The curve of $y = e^x - 2x^2$ is shown on the diagram below. The point P has coordinates $(1, e-2)$.



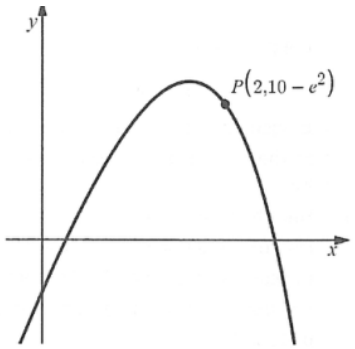
a) Find the equation of the tangent line at P, giving your answer in the form $y = mx + c$ where m and c are constants to be found.
b) Find the coordinates where the tangent line crosses the x-axis and y-axis.
c) Hence calculate the area of the triangle bounded by the tangent line at P and the coordinate axes giving your answer in the form $\dfrac{a}{b-e}$ where a and b are integers to find.

Question 24. A curve has equation $y = 4e^{-kx}$ where k is a positive constant.

a) Find $\dfrac{dy}{dx}$ when $x = 2$ giving your answer in terms of k.
b) Show that $\dfrac{d^2y}{dx^2} = k^2y$.

Question 25. A curve has equation $y = e^{5kx}(1-k)$ where k is a positive constant. Given that at $x = \frac{1}{k}$, $\frac{d^2y}{dx^2} = -100e^5$, find the value of k.

Question 26. The curve of $y = 5x - e^x$ is shown on the diagram below. The point P has coordinates $(2, 10 - e^2)$.



a) Find the equation of the normal line at P, giving your answer in the form $y = mx + c$ where m and c are constants to be found.

b) Find the coordinates where the normal line meet the x-axis, giving your answer in the form $(e^4 + ae^2 + b, 0)$ where a and b are integers to be found.

Question 27. The normal to the curve $f(x) = e^x - 2x^2 - x + e^{-x}$ at $x = 0$ is the same line as the tangent to the curve $g(x) = \frac{1}{2}x^2 + (5 + k)x + p$ at the point $(7, 9)$. Find the values of p and k.